

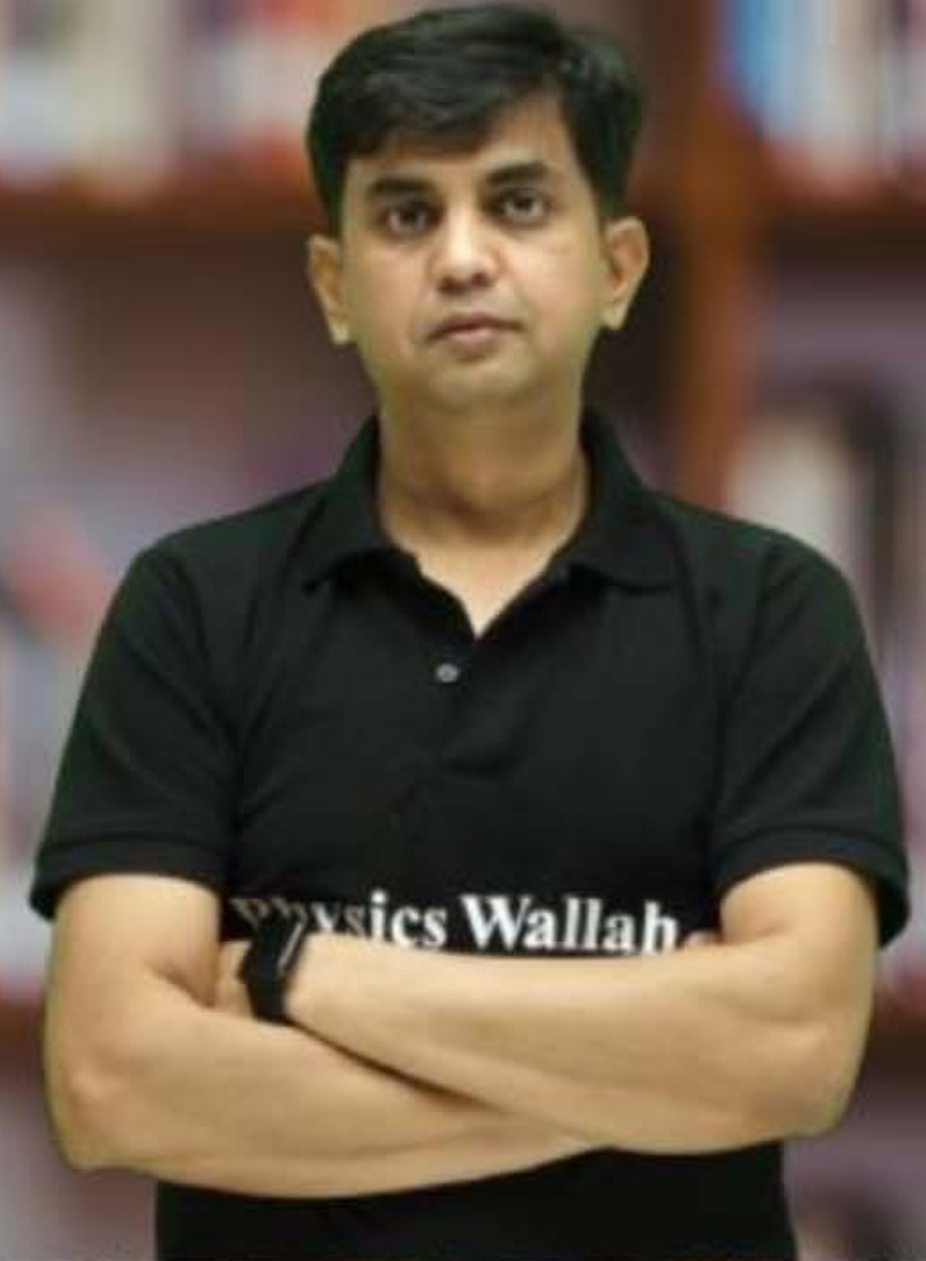
CS & IT ENGINEERING



Computer Network

Flow Control

Lecture No. - 10



By - Abhishek Sir



Recap of Previous Lecture



Topic

End-to-End Delay ✓ $\Rightarrow$ V. Imp.

Topic

Optimal Packet Size



Topics to be Covered



Topic

Framing

ABOUT ME



Hello, I'm **Abhishek**

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- M.Tech (CS) - IIT Kharagpur
- 12 years of GATE CS teaching experience

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#Q. Consider a network path (P—Q—R) between nodes P and R via router Q. Node P sends a file of size 10^6 bytes to R via this path by splitting the file into chunks of 10^3 bytes each. (Node P sends these chunks one after the other without any wait time between the successive chunk transmissions. Assume that the size of extra headers added to these chunks is negligible, and that the chunk size is less than the MTU.

Each of the links P—Q and Q—R has a bandwidth of 10^6 bits/sec, and negligible propagation latency. Router Q immediately transmits every packet it receives from P to R, with negligible processing and queueing delays. Router Q can simultaneously receive on link P—Q and transmit on link Q—R.

Assume P starts transmitting the chunks at time $t = 0$. Which one of the following options gives the time (in seconds, rounded off to 3 decimal places) at which R receives all the chunks of the file?

[GATE 2024, Set-1, 2-Mark]

(A) 8.000

✓ (B) 8.008

(C) 15.992

(D) 16.000

Ans: B

Solution:-

$$\text{Packet Size} = 10^3 \text{ bytes} = 8 * 10^3 \text{ bits}$$

$$\text{Bandwidth} = 10^6 \text{ bits / sec}$$

$$t_x = \frac{\text{Packet Size}}{\text{Bandwidth}} = \frac{8 * 10^3 \text{ bits}}{10^6 \text{ bits / sec}} = 8 * 10^{-3} \text{ sec}$$

$$t_p = 0$$

$$\text{File Size} = 10^6 \text{ bytes}$$

$$\text{Packet Size} = 10^3 \text{ bytes}$$

$$\text{Number of packets (N)} = \frac{\text{File Size}}{\text{Packet Size}} = \frac{10^6 \text{ bytes}}{10^3 \text{ bytes}} = 10^3$$

$$\begin{aligned} \text{End-to-end delay} &= (N * t_x + t_p) + (t_x + t_p) \\ &= (10^3 * 8 * 10^{-3} \text{ sec}) + 8 * 10^{-3} \text{ sec} \\ &= 8.008 \text{ sec} \end{aligned}$$

#Q. Two hosts are connected via a packet switch with 10^7 bits per second links. Each link has a propagation delay of 20 microseconds. The switch begins forwarding a packet 35 microseconds after it receives the same. If 10000 bits of data are to be transmitted between the two hosts using a packet size of 5000 bits, the time elapsed between the transmission of the first bit of data and the reception of the last bit of the data in microseconds is _____.

[GATE 2015, Set-2, 2-Mark]



Solution:-

$$\text{Packet Size} = 5000 \text{ bits} = 5 * 10^3 \text{ bits}$$

$$\text{Bandwidth} = 10^7 \text{ bits / sec}$$

$$t_x = \frac{\text{Packet Size}}{\text{Bandwidth}} = \frac{5 * 10^3 \text{ bits}}{10^7 \text{ bits / sec}} = 500 \mu\text{s}$$

$$t_{p_1} = 20 \mu\text{s} = t_{p_2}$$

$$\text{Processing Delay} = 35 \mu\text{s}$$

$$\text{File Size} = 10,000 \text{ bits}$$

$$\text{Packet Size} = 5,000 \text{ bits}$$

$$\text{Number of packets (N)} = \frac{\text{File Size}}{\text{Packet Size}} = \frac{10000 \text{ bits}}{5000 \text{ bits}} = 2$$

$$\begin{aligned} \text{End-to-end delay} &= (N * t_x + t_p) + \text{Processing Delay} + (t_x + t_p) \\ &= (2 * 500 + 20) + 35 + (500 + 20) \mu s \\ &= 1575 \mu s \end{aligned}$$

#Q. Consider a source computer (S) transmitting a file of size 10^6 bits to a destination computer (D) over a network of two routers (R1 and R2) and three links (L1, L2, and L3). L1 connects S to R1; L2 connects R1 to R2; and L3 connects R2 to D. Let each link be of length 100 km. Assume signals travel over each link at a speed of 10^8 meters per second. Assume that the link bandwidth on each link is 1Mbps. Let the file be broken down into 1000 packets each of size 1000 bits. Find the total sum of transmission and propagation delays in transmitting the file from S to D.

[GATE 2012, 2-Mark]



☒ (A) 1005 ms
(C) 3000 ms

(B) 1010 ms
(D) 3003 ms

Ans: A

Solution:-

$$\text{Packet Size} = 1000 \text{ bits} = 10^3 \text{ bits}$$

$$\text{Bandwidth} = 1 \text{ Mbps} = 10^6 \text{ bits / sec}$$

$$t_x = \frac{\text{Packet Size}}{\text{Bandwidth}} = \frac{10^3 \text{ bits}}{10^6 \text{ bits / sec}} = 1 \text{ ms}$$

$$\text{Distance} = 100 \text{ Km} = 10^5 \text{ m}$$

$$\text{Signal Speed} = 10^8 \text{ m/s}$$

$$t_p = \frac{\text{Distance}}{\text{Signal Speed}} = \frac{10^5 \text{ m}}{10^8 \text{ m/s}} = 1 \text{ ms} = t_{P2} = t_{P3}$$

$$\text{File Size} = 10^6 \text{ bits}$$

$$\text{Packet Size} = 10^3 \text{ bits}$$

$$\text{Number of packets (N)} = \frac{\text{File Size}}{\text{Packet Size}} = \frac{10^6 \text{ bits}}{10^3 \text{ bits}} = 10^3$$

$$\begin{aligned} \text{End-to-end delay} &= (N * t_x + t_p) + (t_x + t_p) + (t_x + t_p) \\ &= (10^3 * 1 + 1) + (1 + 1) + (1 + 1) \text{ ms} \\ &= 1005 \text{ ms} \end{aligned}$$



Topic : Framing

⇒ Frame Synchronization



Problem : How receiver identify frame boundaries while receiving multiples frames?

[Variable length frames and transmitted without time-gap]



Topic : Framing





Topic : Framing



Problem : How receiver identify **frame boundaries** while receiving **multiples frames**?

[Variable length frames and transmitted without time-gap]

Solution :

1. Byte (Character) Count
2. Byte (Character) Stuffing
3. Bit Stuffing (CS-04, CS-14)

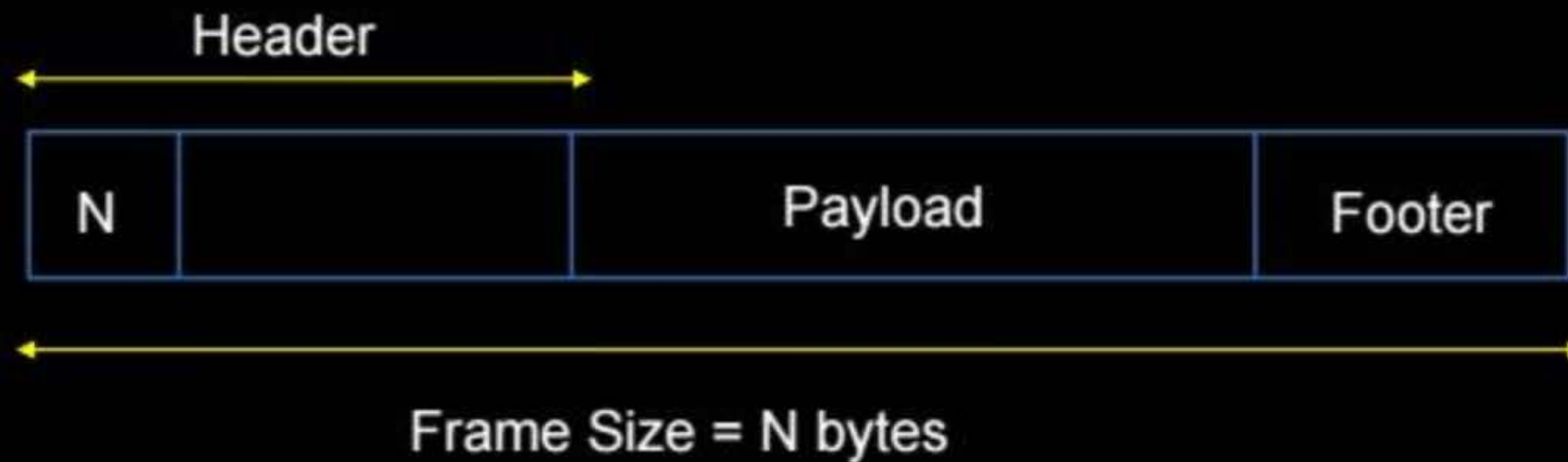


Topic : Byte Count



Byte Count field :

- One or two byte in size
- At the begin of the frame [Part of Header]
- Contains length of frame in bytes
[Size of Byte Count field inclusive]





Topic : Byte Count





Topic : Byte Count



Disadvantage :

if error occurred in the “Byte Count field”

then entire transmission is garbled [that frame onwards]

[Need to retransmit more than one frame]



Topic : Byte Stuffing



Special characters :

1. **"STX"** : [Start of Text / Transmission]
Start Frame Delimiter (SFD), ASCII value = 2
2. **"ETX"** : [End of Text / Transmission]
End Frame Delimiter (EFD), ASCII value = 3
3. **"ESC"** : [Escape character]
ASCII value = 27



Topic : Byte Stuffing

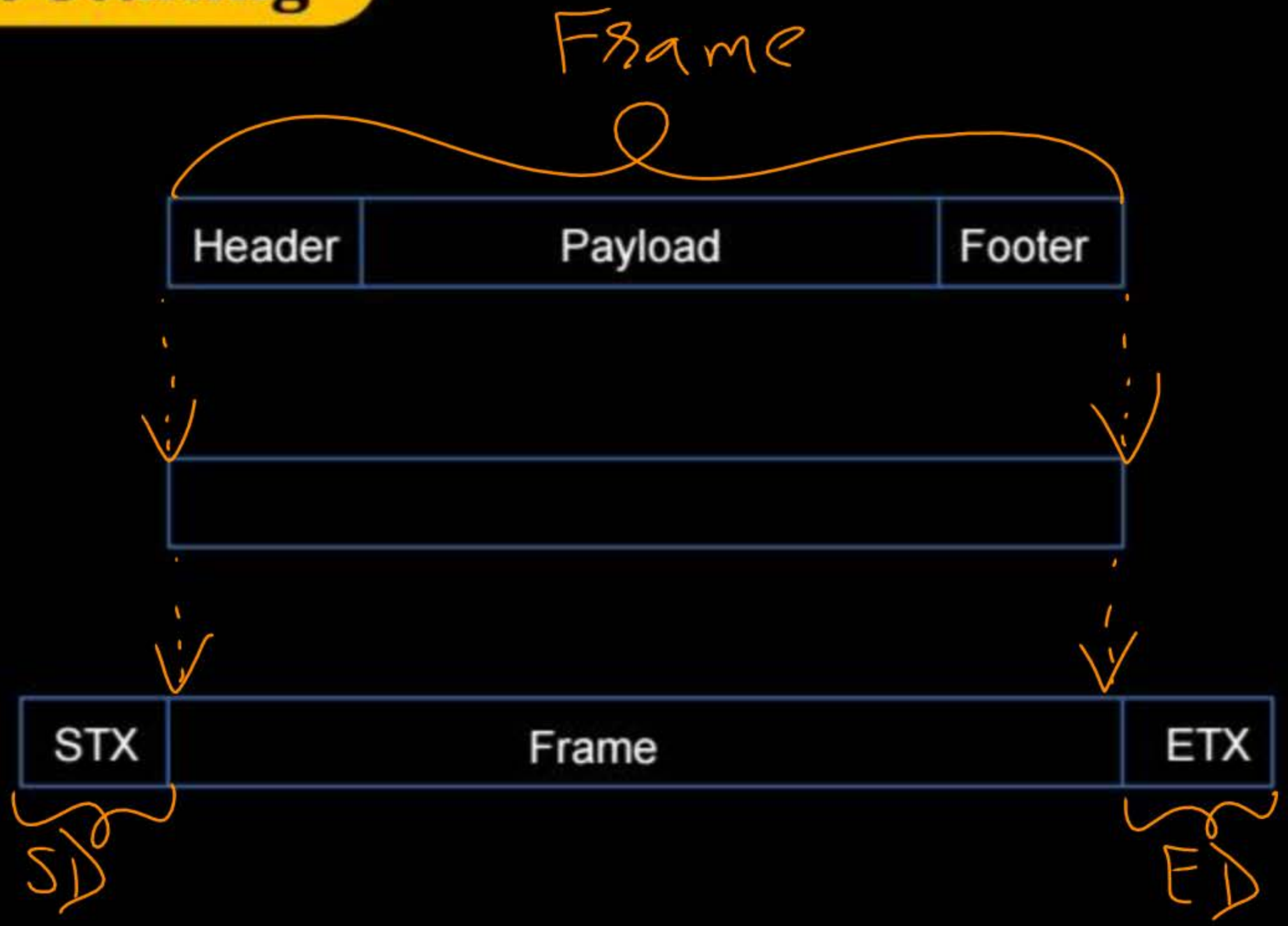


Transmitter Protocol :

1. Transmit **“STX”** just before frame transmission start.
2. Transmit **“ETX”** just after frame transmission completed.
3. Stuff **“ESC”** character just before every **“STX” / “ETX” / “ESC”** character
appear in data while transmission. [Except start and end of frame]

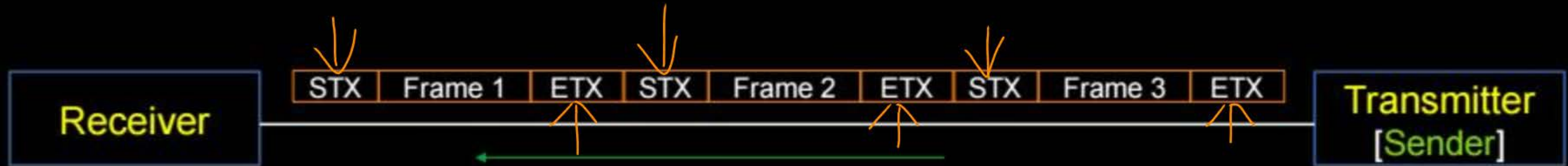


Topic : Byte Stuffing





Topic : Byte Stuffing





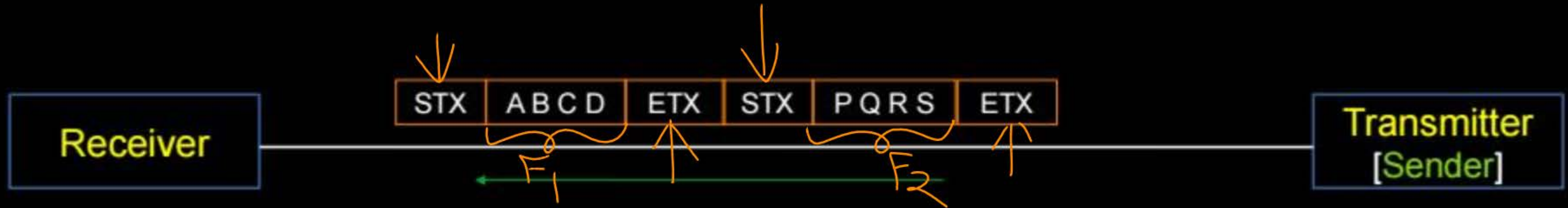
Topic : Byte Stuffing



CASE I :

Frame 1 : "A B C D"

Frame 2 : "P Q R S"





Topic : Byte Stuffing



CASE II :

Frame 1 : "A B **STX** D"

Frame 2 : "P Q **ESC** S"





Topic : Byte Stuffing

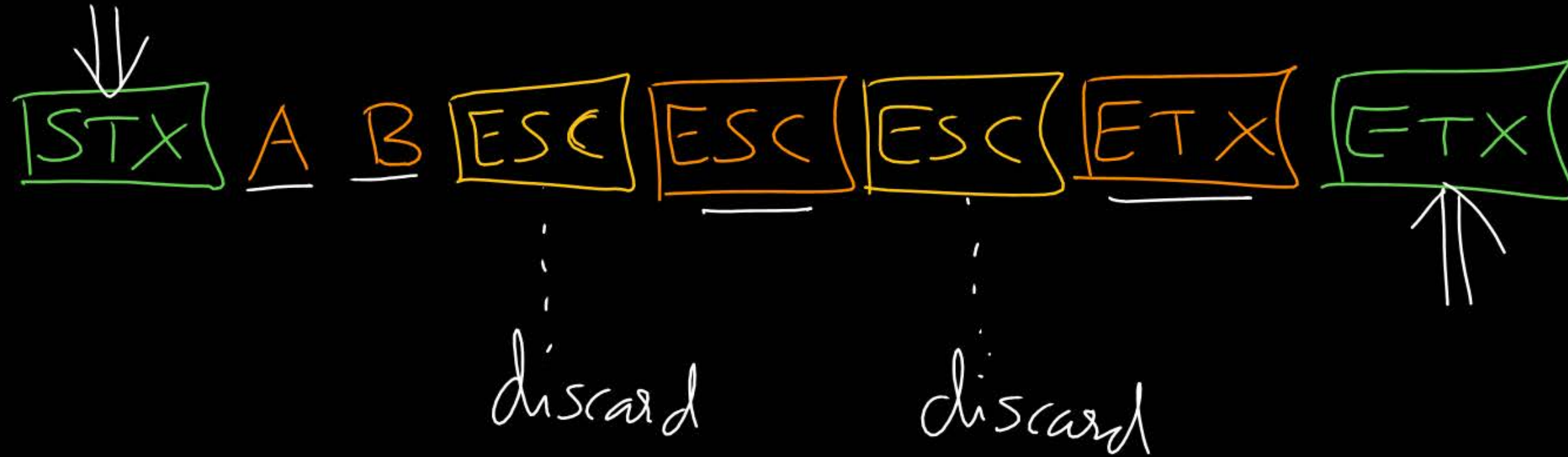


Receiver Protocol :

1. Looking for **"STX"** (start of frame)
2. After **"STX"** looking for **"ETX"** (End of frame)
3. Discard **"ESC"** character while receiving the data frame and copy next character into the receiving buffer

Frame = A B ESC ETX

Transmitted data =





Topic : Byte Stuffing



Disadvantage :

Every **“STX” / “ETX” / “ESC”** (special character) in the data frame will increase length of frame by **“one-byte”**.

[Except Start and End of Frame]



Topic : Bit Stuffing



Replace "**STX**" and "**ETX**" with "**FLAG**" character.

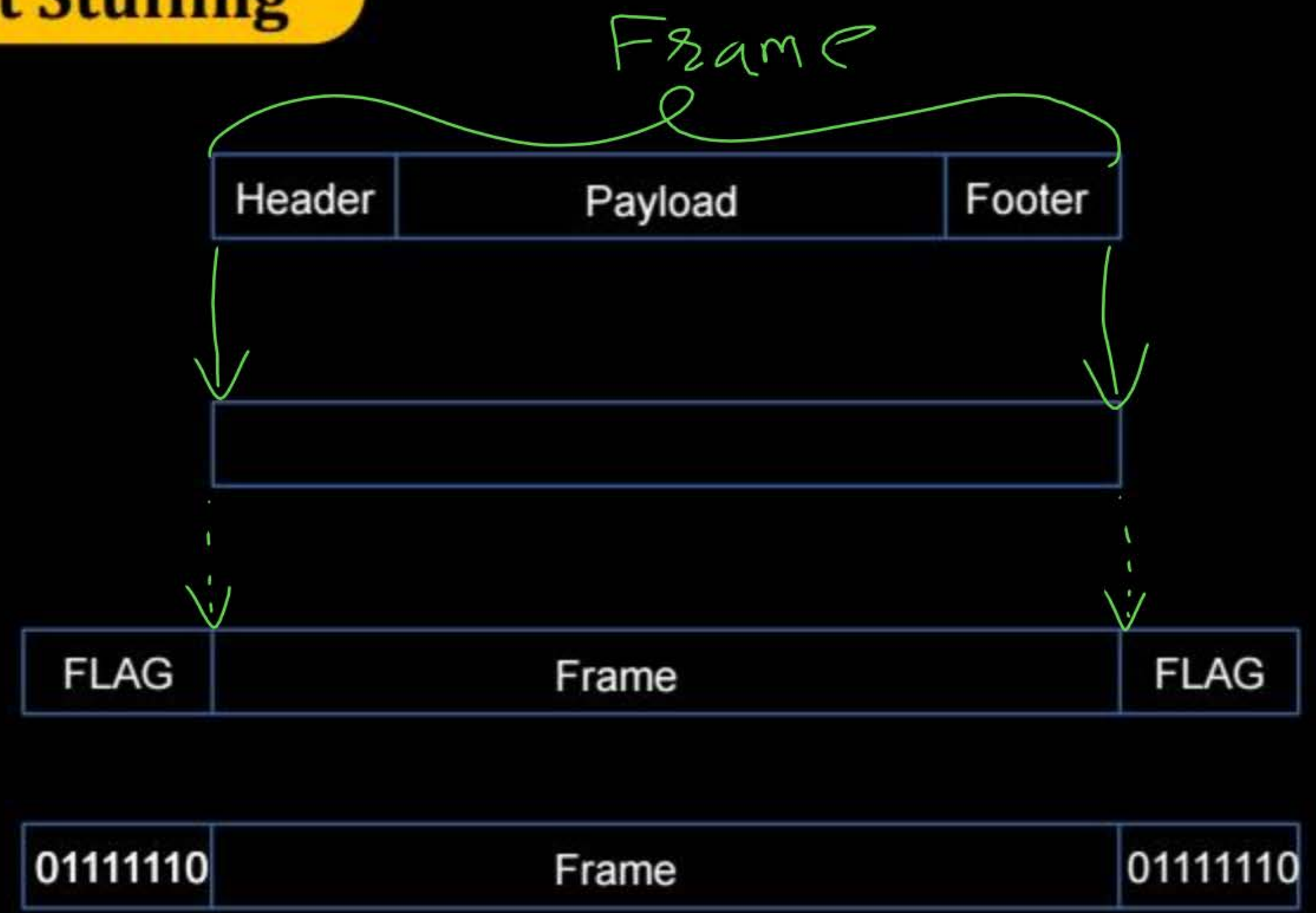
FLAG : "~"

Binary (bit pattern) = "01111110"

ASCII Value = 126 = 0x7E



Topic : Bit Stuffing





Topic : Bit Stuffing



* Two consecutive Flag, shows end of one frame and start of next frame.



Topic : Bit Stuffing



Transmitter Protocol :

1. Transmit **“FLAG”** character, just before frame transmission start.
2. Transmit **“FLAG”** character, just after frame transmission completed.
3. Stuff **“Zero Bit”** after every **five continuous one's** while transmission.
[Except start and end of frame]



Topic : Bit Stuffing



Receiver Protocol :

1. Looking for "Start of Frame" **[FLAG]**.
2. After Start of Frame, it is looking for "End of Frame" **[FLAG]**.
3. Discard "Zero Bit" that found after every five continuous one's while receiving data frame.



Topic : Bit Stuffing



CASE I :

Frame 1 = 11011101110101

Output = 01111110 11011101110101 01111110

FLAG START Frame FLAG ENG



Output = 01111110 1 0 1 1 1 1 1 0 1 0 1 0 1 01111110

START FLAG

1 2 3 4 5

stuff bit



CASE III :

Frame 3 = 10111101010
 1 2 3 4 5

Output = 01111110 1 0 1 1 1 1 1 0 0 1 0 1 0 01111110

START FLAG

1 2 3 4 5

stuff bit



Topic : Bit Stuffing



CASE IV :

Frame 4 = 111110111101
 1 2 3 4 5

Output = 01111110 1 1 1 1 1 0 1 1 1 1 0 1 01111110
 Start FLAG 1 2 3 4 5 ↑ End Flag
 stuf

FLAG = 011110

Data = 1110 1111 1110 1110
 1 2 3 1 2 3 1 2 3 1 2 3

Transmitted Data =

011110 1110 01110 1110 01110 01110 001110



Topic : Bit Stuffing



Conclusion :

Six continuous one's can only appear in the start and end of frame while transmission.

Advantage :

For every five continuous one's present in the data frame, length of frame increases by "one-bit".

#Q. A bit-stuffing based framing protocol uses an 8-bit delimiter pattern of 01111110. If the output bit-string after stuffing is 01111100101, then the input bit-string is

- (A) 0111110100
- (B) 0111110101
- (C) 0111111101
- (D) 0111111111

[GATE-2014]

IIT-KGP
H.W

01110

#Q. In a data link protocol, the frame delimiter flag is given by 0111. Assuming that bit stuffing is employed, the transmitter sends the data sequence 01110110 as

[GATE-2004]

- (A) 01101011
- (B) 011010110
- (C) 011101100
- (D) 0110101100

11T-D
[H.W.]



2 mins Summary



Topic

Byte Count

Topic

Byte Stuffing

Topic

Bit Stuffing ✓



THANK - YOU

